

Changes: §3, §7.2.1, §7.2.2

Deletions: §3: LIAISON® EASY Wash/System Liquid (REF 319301)

LIAISON® NPAT-2 (REF 319123)

1. INTENDED USE

The LIAISON® NPAT-2 assay uses chemiluminescence technology to check LIAISON® XL and LIAISON® XS Analyzers performance in the following instances:

1. As a part of Final Test Protocol at the end of instrument manufacturing process;
2. During periodical maintenance of the instrument at the end-user's site (as needed).

The kit should be used by skilled personnel only. Use should be further restricted to the two following areas:

1. Personnel at the instrument manufacturer's site;
2. DiaSorin field service engineers;
3. Field service engineers qualified by DiaSorin.

With the purpose of verifying compliance of the following parameters with expected performance:

1. Repeatable reagent dispensing (precision);
2. Washer performance (precision and yield, curbing of non-specific binding or liquid splashes).

2. DESCRIPTION

The LIAISON® NPAT-2 kit is used in three different assay protocols combined in a Combi-assay structure (**displayed as NPAT-2 on the analyzer**), as follows:

1. Protocol one (**displayed as MP1S1C on the analyzer**): Basic evaluation of correct right pipettor and washer performance, based upon dispensing of ABEI-conjugated magnetic particles
2. Protocol two (**displayed as MP3S1C on the analyzer**): Evaluation of correct performance and yield of the washer for particle resuspension;
3. Protocol three (**displayed as W100 on the analyzer**): Evaluation of the washer ability to effectively remove unbound tracer and splashes from the cuvette during washing.

3. MATERIALS PROVIDED

Reagent integral

Magnetic particles (2.5 mL)	SORB	Magnetic Particles coated with a protein conjugated to an isoluminol derivative in a buffer with surfactants and less than 0.1% sodium azide.
Conjugate (4.3 mL)	CONJ	Mouse monoclonal antibody conjugated to an isoluminol derivative in phosphate buffer, BSA and 0.2% ProClin® 300.
Buffer (3 x 28 mL)	BUF	Phosphate buffer solution with surfactants, BSA and 0.2% ProClin® 300.

All reagents are supplied ready to use. The order of reagents reflects the layout of containers in the reagent integral.

Overall number of tests available is 610:

For LIAISON® XL, they are allocated for each of the three protocols as follows:

1. MP1S1C (protocol one): 55 tests;
2. MP3S1C (protocol two): 55 tests;
3. W100 (protocol three): 500 tests.

For LIAISON® XS, they are allocated for each of the three protocols as follows:

1. MP1S1C (protocol one): 45 tests;
2. MP3S1C (protocol two): 45 tests;
3. W100 (protocol three): 150 tests.

Materials required but not provided

LIAISON® XL Analyzer	LIAISON® XS Analyzer
LIAISON® XL Cuvettes (REF X0016). LIAISON® XL Disposable Tips (REF X0015) or LIAISON® Disposable Tips (REF X0055). LIAISON® XL Starter Kit (REF 319200). LIAISON® EASY Starter Kit (REF 319300). LIAISON® Wash/System Liquid (REF 319100). LIAISON® XL Waste Bags (REF X0025).	LIAISON® Cuvettes on Tray (REF X0053). LIAISON® Disposable Tips (REF X0055). LIAISON® EASY Starter Kit (REF 319300). LIAISON® EASY Waste (REF X0054). LIAISON® EASY System Liquid (REF 319302). LIAISON® EASY Wash Buffer (REF 319301). LIAISON® EASY Cleaning Tool (REF 310996).

4. SAFETY PRECAUTIONS

Do not eat, drink, smoke or apply cosmetics in the assay laboratory.

Do not pipette by mouth.

Avoid direct contact with potentially infected material by wearing laboratory clothing, protective goggles, and disposable gloves. Wash hands thoroughly at the end of each assay.

Avoid splashing or forming an aerosol. All drops of biological reagent must be removed with a sodium hypochlorite solution with 0.5% active chlorine, and the means used must be treated as infected waste.

All samples and reagents containing biological materials used for the assay must be considered as potentially able to transmit infectious agents. The waste must be handled with care and disposed of in compliance with the laboratory guidelines and the statutory provisions in force in each Country. Any materials for reuse must be appropriately sterilized in compliance with the local laws and guidelines. Check the effectiveness of the sterilization/decontamination cycle.

Do not use kits or components beyond the expiration date given on the label.

Reagents containing sodium azide (< 0.1%) [EC no: 247-852-1]:

DIRECTIVE	EC No. 1272/2008
HAZARD / RISK STATEMENTS	EUH 210 - Safety data sheet available on request

Reagents containing ProClin®:

DIRECTIVE	EC No. 1272/2008
REAGENTS	[CONJ] [BUF]
CLASSIFICATION OF SUBSTANCE	Skin sensitizer, Category 1
SIGNAL WORD	Warning!
SYMBOLS / PICTOGRAMS	 GHS07 -Exclamation mark
HAZARD / RISK STATEMENTS	H317 - May cause an allergic skin reaction.
PRECAUTIONARY / SAFETY STATEMENTS	P261 - Avoid breathing mist or spray. P280 - Wear protective gloves and clothing and eye protection. P363 - Wash contaminated clothing before reuse
CONTAINS: (only substances prescribed pursuant to Article 18 of EC Regulation 1272/2008).	reaction mass of: 5-chloro-2-methyl-4-isothiazolin-3-one [EC no. 247-500-7] and 2-methyl-2H -isothiazol-3-one [EC no. 220-239-6] (3:1) (ProClin® 300).

For additional information see Safety Data Sheets available on www.diasorin.com.

5. PREPARATION OF REAGENT INTEGRAL

Before removing the seals from the containers, gently and carefully shake the reagent integral horizontally. Avoid formation of foam. Remove the seal from each container and turn the thumb wheel at the bottom of the magnetic particle container to and fro until the suspension turns brown. This procedure initiates resuspension of magnetic particles. Carefully wipe the surface of each septum to remove residual liquid. Then operate as follows.

LIAISON® XL and LIAISON® XS Analyzers are equipped with a built-in solid-state magnetic device which aids in the dispersal of microparticles prior to placement of a reagent integral into the reagent area of the analyzer (refer to the analyzer operator's manuals for details).

Insert the reagent integral into the dedicated slot.

Allow the reagent integral to remain in the device for at least 30 seconds. Repeat as necessary.

Place the integral into the reagent area of the analyzer with the label facing left and let it stand for 15 minutes before using. The analyzer automatically stirs and completely resuspends the magnetic particles.

Follow the analyzer operator's manuals to load the specimens and start the run.

6. REAGENT INTEGRAL STORAGE AND STABILITY

Upon receipt, the reagent integral must be stored in an upright position to facilitate resuspension of magnetic particles.

When the reagent integral is stored sealed and kept upright, the reagents are stable at 2-8°C up to the expiry date. Do not freeze. The reagent integral should not be used past the expiry date indicated on the kit and reagent integral labels. After removing the seals, the reagent integral is stable for one day refrigerated either at 2-8°C or on board the instrument.

7. ASSAY PROCEDURE

7.1. Performed on the LIAISON® XL Analyzer

The procedure should be run stepwise in the given order, allowing the system to complete each test before starting the next.

All protocols shall be run on the same integral within the same day.

The yield shall be evaluated based on data obtained from MP1S1C and MP3S1C performed on the same integral within the same day.

Integral shall not be exchanged between LIAISON® XL and LIAISON® XS Instruments.

7.1.1. Assessment of reagent dispensing and washing assessment

7.1.1.1. MP1S1C (protocol one): Basic performance of right pipettor and washer

Assay protocol to be used is MP1S1C.

Integral for testing is to be loaded in lane 9. Allow 15 minutes for particle re-suspension.

The test should be run in 55 replicates, using a series of five tubes filled with working Wash/System liquid used as a sample. Each tube must be run in 11 replicates (total volume used per tube: 110 µL plus dead volume), according to the following protocol:

1. Dispense NPAT-2 buffer for particle dilution and 20 µL of magnetic particles;
2. Dispense sample;
3. Incubate;
4. Wash and separate the particles with five washing step;
5. Add starter reagents and read signal emitted;
6. Calculate mean, standard deviation and percent CV of the signal;
7. Check whether %CV meets acceptance criteria.

7.1.1.2. MP3S1C (protocol two): Washer performance and yield

Assay protocol to be used is MP3S1C.

Integral for testing is to be loaded in lane 9. Allow 15 minutes for particle re-suspension.

The test should be run in 55 replicates, using a series of five tubes filled with working Wash/System liquid used as a sample. Each tube must be run in 11 replicates (total volume used per tube: 110 µL plus dead volume), according to the following protocol:

1. Dispense NPAT-2 buffer for particle dilution and 20 µL of magnetic particles;
2. Dispense sample;
3. Incubate;
4. Wash and separate the particles with five washing step;
5. Dispense NPAT-2 buffer for particle dilution;
6. Incubate;
7. Wash and separate the particles with five washing step;
8. Dispense NPAT-2 buffer for particle dilution;
9. Incubate;
10. Wash and separate the particles with five washing step;
11. Add starter reagents and read signal emitted;
12. Calculate mean, standard deviation and percent CV of the signal;
13. Calculate the ratio of test 2 mean signal to test 1 mean signal (yield of the washer);
14. Check whether %CV meets acceptance criteria.

7.1.2. Assessment of washer performance and removal of non-specific binding and splashes assessment

7.1.2.1. W100 (protocol three)

Assay protocol to be used is W100.

Integral for testing is to be loaded in lane 9.

The test should be run in 500 replicates, using a series of 25 tubes filled with working Wash/System liquid used as a sample. Each tube must be run in 20 replicates (total volume used per tube: 200 µL plus dead volume), according to the following protocol:

1. Dispense conjugate;
2. Incubate;
3. Wash cuvettes with five washing step;

4. Add starter reagents and read signal emitted;
5. Calculate mean, standard deviation and percent CV of the signal;
6. Calculate number of outliers in the overall result, as the number of results whose signal is equal to or greater than 1.5 times the mean signal observed;
7. Check whether signal and number of outliers meet acceptance criteria.

7.2. Performed on the LIAISON® XS Analyzer

The procedure should be run stepwise in the given order, allowing the system to complete each test before starting the next.

All protocols shall be run on the same integral within the same day.

The yield shall be evaluated based on data obtained from MP1S1C and MP3S1C performed on the same integral within the same day.

Integral shall not be exchanged between LIAISON® XL and LIAISON® XS Instruments.

7.2.1. Assessment of reagent dispensing and washing assessment

7.2.1.1 MP1S1C (protocol one): Basic performance of right pipettor and washer

Assay protocol to be used is MP1S1C.

Allow 15 minutes for particle re-suspension.

The test should be run in 45 replicates, using a series of five tubes filled with working [EASY Wash Buffer](#) used as a sample. Each tube must be run in 9 replicates (total volume used per tube: 110 µL plus dead volume), according to the following protocol:

1. Dispense NPAT-2 buffer for particle dilution and 20 µL of magnetic particles;
2. Dispense sample;
3. Incubate;
4. Wash and separate the particles with five washing step;
5. Add starter reagents and read signal emitted;
6. Calculate mean, standard deviation and percent CV of the signal;
7. Check whether %CV meets acceptance criteria.

7.2.1.2 MP3S1C (protocol two): Washer performance and yield

Assay protocol to be used is MP3S1C.

Allow 15 minutes for particle re-suspension.

The test should be run in 45 replicates, using a series of five tubes filled with working [EASY Wash Buffer](#) as a sample. Each tube must be run in 9 replicates (total volume used per tube: 110 µL plus dead volume), according to the following protocol:

1. Dispense NPAT-2 buffer for particle dilution and 20 µL of magnetic particles;
2. Dispense sample;
3. Incubate;
4. Wash and separate the particles with five washing step;
5. Dispense NPAT-2 buffer for particle dilution;
6. Incubate;
7. Wash and separate the particles with five washing step;
8. Dispense NPAT-2 buffer for particle dilution;
9. Incubate;
10. Wash and separate the particles with five washing step;
11. Add starter reagents and read signal emitted;
12. Calculate mean, standard deviation and percent CV of the signal;
13. Calculate the ratio of test 2 mean signal to test 1 mean signal (yield of the washer);
14. Check whether %CV meets acceptance criteria.

7.2.2 Assessment of washer performance and removal of non-specific binding and splashes assessment

7.2.2.1 W100 (protocol three)

Assay protocol to be used is W100.

W100 can be repeated on the same integral within the same day at maximum 3 times.

The test should be run in 150 replicates, using a series of 15 tubes filled with working [EASY Wash Buffer](#) used as a sample. Each tube must be run in 10 replicates (total volume used per tube: 200 µL plus dead volume), according to the following protocol:

1. Dispense conjugate;
2. Incubate;
3. Wash cuvettes with five washing step;
4. Add starter reagents and read signal emitted;
5. Calculate mean, standard deviation and percent CV of the signal;
6. Calculate number of outliers in the overall result, as the number of results whose signal is equal to or greater than 1.5 times the mean signal observed;
7. Check whether signal and number of outliers meet acceptance criteria.

8. ACCEPTANCE CRITERIA

The instrument is suitable for use if acceptance criteria are met.

Refer to Corporate Service specific documentation for reference values.